

Case Study: Car Black Box using STM32F407

Introduction

A Car Black Box (Event Data Recorder) is an embedded system that records critical vehicle information, driver actions, and contextual data before, during, and after an event (e.g., collision). It provides valuable evidence for accident analysis, insurance claims, fleet management, and improving vehicle safety.

Use Cases

- Accident investigation – analyze cause of crashes.
- Insurance claim validation – provide factual evidence.
- Driver behavior monitoring – track speed, braking, acceleration.
- Fleet management – monitor and optimize vehicle usage.
- Legal evidence – support in court cases.

Features

- Event data recording (speed, acceleration, braking, impact).
- Timestamp logging using RTC (internal or external).
- Storage on SD card with FATFS file system.
- Sensor interfacing (accelerometer, gyroscope, etc.).
- GPS integration for location and time sync.
- Communication via UART/USB for data retrieval and config.
- Power-failure protection with backup capacitor or small battery.

Block Diagram (Text Representation)

Input Modules:

- Sensors: Accelerometer, Gyroscope
- Buttons / Vehicle CAN (optional)
- Power Sources: Vehicle battery (main), Backup capacitor/small battery

Core (STM32F407 MCU):

- Peripherals: I2C, SPI, ADC, SDIO, UART
- RTC: Internal LSE-based RTC or external (DS3231)
- Functions: Event detection, timestamping, file management (FATFS), power-fail handling

Output / Interfaces:

- SD Card (FAT filesystem)
- GPS Module (UART)
- UART / USB for data retrieval
- LEDs for status indication
- External Systems: Fleet management servers, Diagnostic tools

Notes

- The STM32F407 acts as the central processing unit, collecting data from inputs, timestamping events using RTC, and storing logs on the SD card using FATFS.
- GPS provides location/time reference.
- LEDs and buttons provide simple user interaction.
- UART/USB provide a way to extract or view logs.

